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Object Oriented Programming Using Java

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Objectives

- Compile time polymorphism/Static binding.

Static Polymorphism

- The compiler resolves the polymorphism during the compilation of the program. Also, called static binding.
- Achieved by method overloading.
- Defining two or more methods within the class with same name.
- Method invoked is based on the reference type.

Method Overloading

- Process of defining two or more methods within the same class with same name and with different parameter declarations.
- Methods are said to be overloaded.
- Method overloading is one of Java's most exciting and useful features.
- Java uses the type and/or number of arguments to determine which version of the overloaded method to actually call.

Example Program

```
class Overload {
    void display() {
        System.out.println("No parameters");
    }

    // Overload display for one integer parameter.
    void display(int a) {
        System.out.println("a: " + a);
    }

    void display(int a, int b) {
        System.out.println("a and b: " + a + " " + b);
    }

    // Overload display for a double parameter
    double display(double a) {
        System.out.println("double a: " + a);
        return a * a;
    }
}
```

Example Program

```
class Example {
    public static void main(String args[]) {
        Overload obj = new Overload();
        double result;

        // Call all versions of display()
        obj.display();
        obj.display(10);
        obj.display(10, 20);
        result = obj.display(123.25);

        System.out.println("Result = " + result);
    }
}
```

Output:

No parameters
a: 10
a and b: 10 20
double a: 123.25
Result =15190.5625

Example Program

- display() is the overloaded function.
- When an overloaded method is called, Java looks for a match between the arguments used to call the method and the method's parameters.
- This match need not always be exact.
- In some cases, Java's automatic type conversions can play a role in overload resolution.

Example Program

```
class Example {
    void display(double num) {
        System.out.println("Display with double: " + num);
    }
}

public class Main {
    public static void main(String[] args) {
        Example obj = new Example();

        obj.display(42);      // Calls display(double num)
        obj.display(3.14);    // Calls display(double num)
    }
}
```

Example Program

- Suppose display() function is not having a single integer parameter here(Avoid void display(int a)).
- If we call obj.display(60) then display(double) will be invoked.
- Java can automatically convert an integer into a double, and this conversion can be used to resolve the call.

Advantages of method Overloading

- Method overloading is one method java implements "one interface and multiple methods" paradigm.
- In languages that do not support method overloading, each method must be given a unique name.
- Frequently if you want to implement essentially the same method for different types of data then method overloading helps.

  abs() function

Advantages of method Overloading

- The value of overloading is that it allows related methods to be accessed by use of a common name.
- Through the application of polymorphism, several names have been reduced to one.
- Though there is no rule that we need to overload related methods it's a good practice to overload closely related operations.

Takeaways

- Compile time polymorphism/static binding.

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Reference

[1] Naughton P. and Schildt H., “Java2 Complete Reference”, Eighth Edition, Tata McGraw- Hill, 2011.

[2] Ali Bahrami, “Object Oriented Systems Development”, Second Edition, McGraw-Hill, 2008